

GEV fit complete.

Datasets / Dallas – TMIN / GEV: Dallas TMIN GEV

Dallas TMIN GEV

Success · created 2026-05-17 01:22 · finished 2026-05-17 01:22

Run again

Delete

Years	Location (μ)	Scale (σ)	Shape (ξ)	Log-likelihood
86	17.578	5.565	-0.1852	-274.32
1940–2025	°F	°F	Weibull (bounded)	converged

REFERENCE EXCEEDANCE

P(annual min ≥ 13.10 °F in any given year) = 0.8796 · expected return period ≈ 1.1 years (reference: 2025 observed maximum)

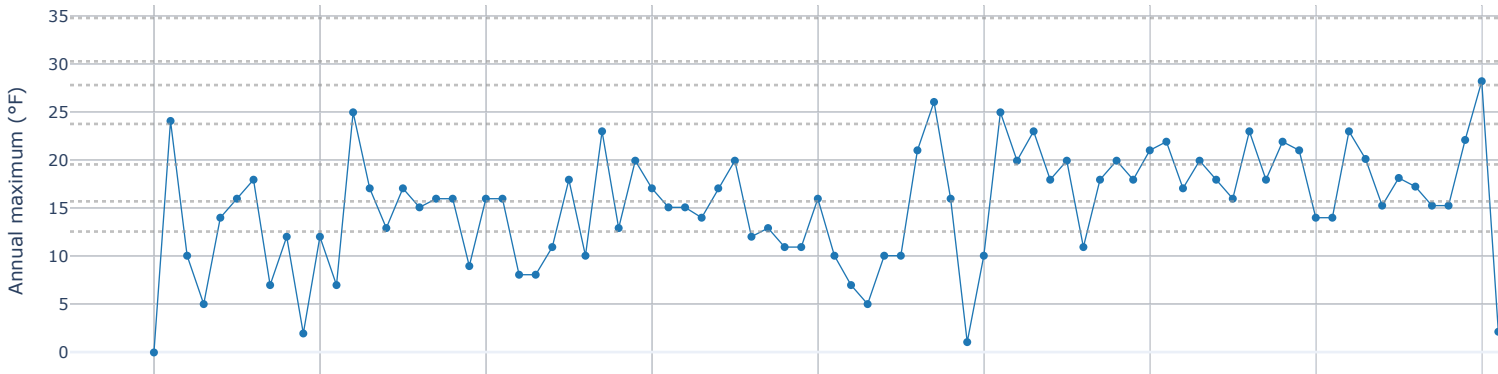
95% CONFIDENCE INTERVALS

Method	Iterations	Successful refits	Location (μ)	Scale (σ)	Shape (ξ)
Parametric	500	500	[16.381, 18.863] SD 0.635	[4.542, 6.524] SD 0.500	[-0.3398, -0.0419] SD 0.0758
Nonparametric	500	500	[12.079, 14.670] SD 0.692	[5.187, 7.202] SD 0.509	[-0.5450, -0.2701] SD 0.0685

RETURN LEVELS

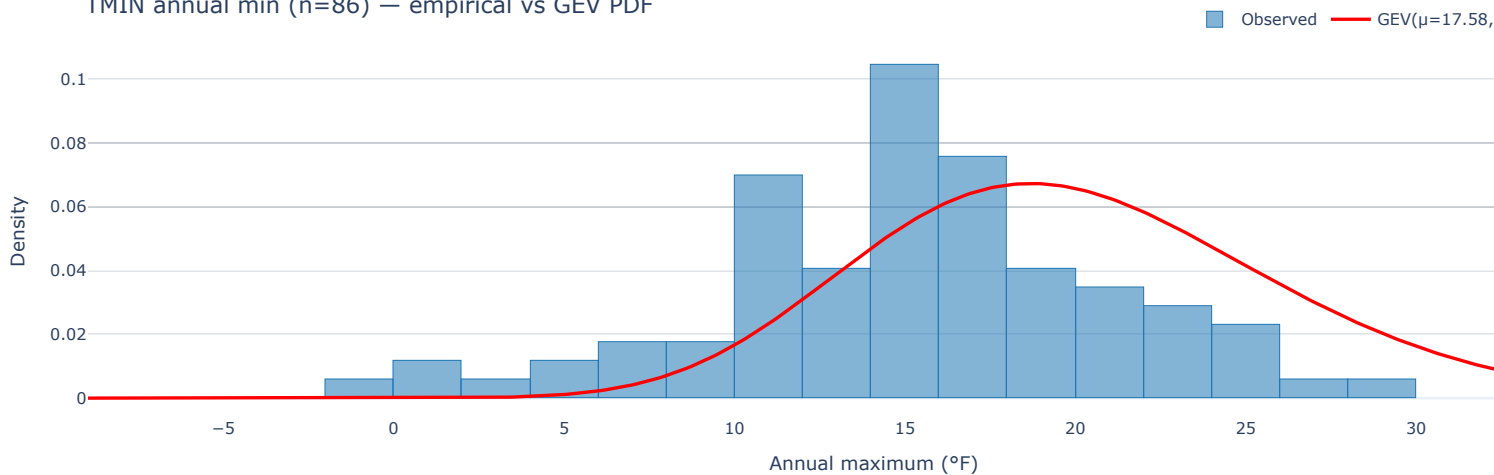
Return period (years)	Return level (°F)
2	19.550
5	24.866
10	27.819
25	31.008
50	33.038
100	34.807
200	36.356
500	38.118
1000	39.264

TMIN annual min (n=86) with fitted percentiles

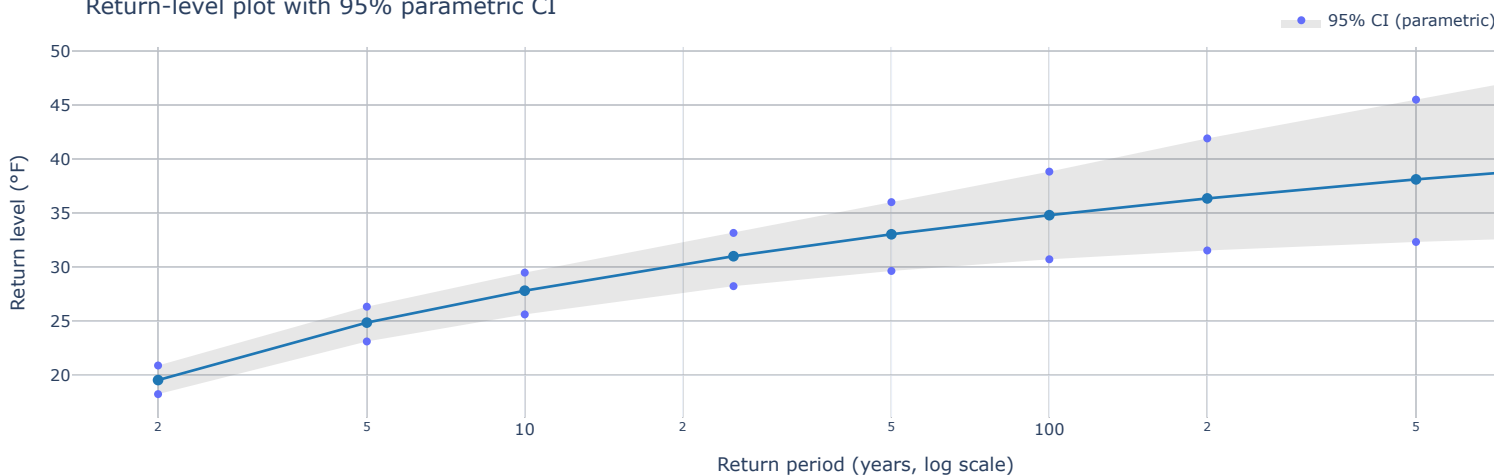


Year

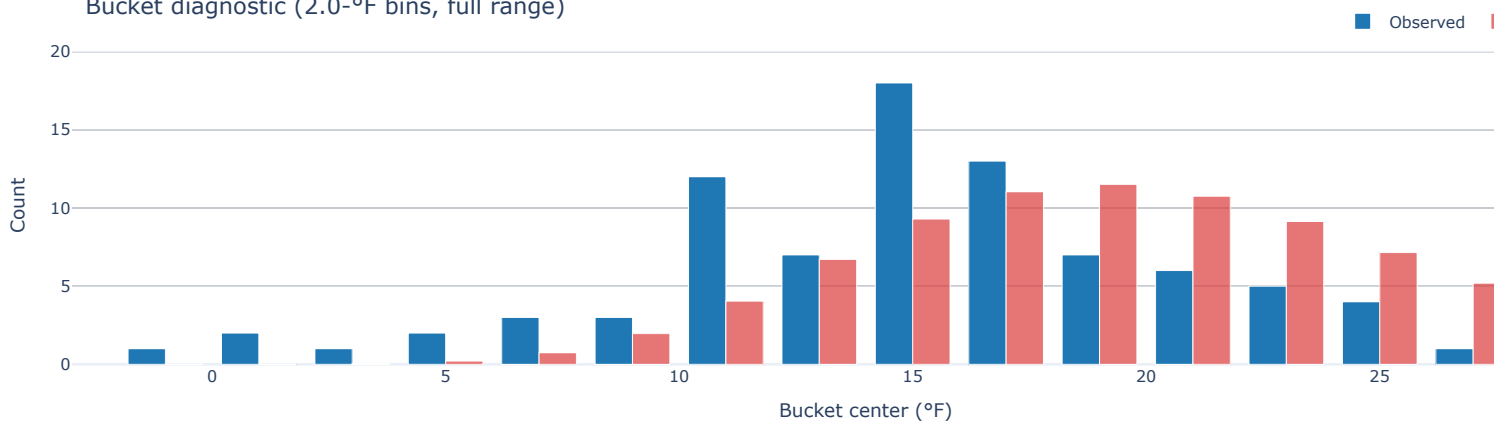
TMIN annual min (n=86) — empirical vs GEV PDF



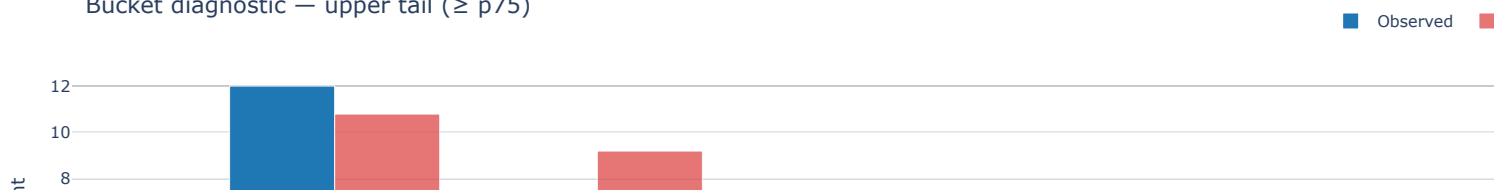
Return-level plot with 95% parametric CI

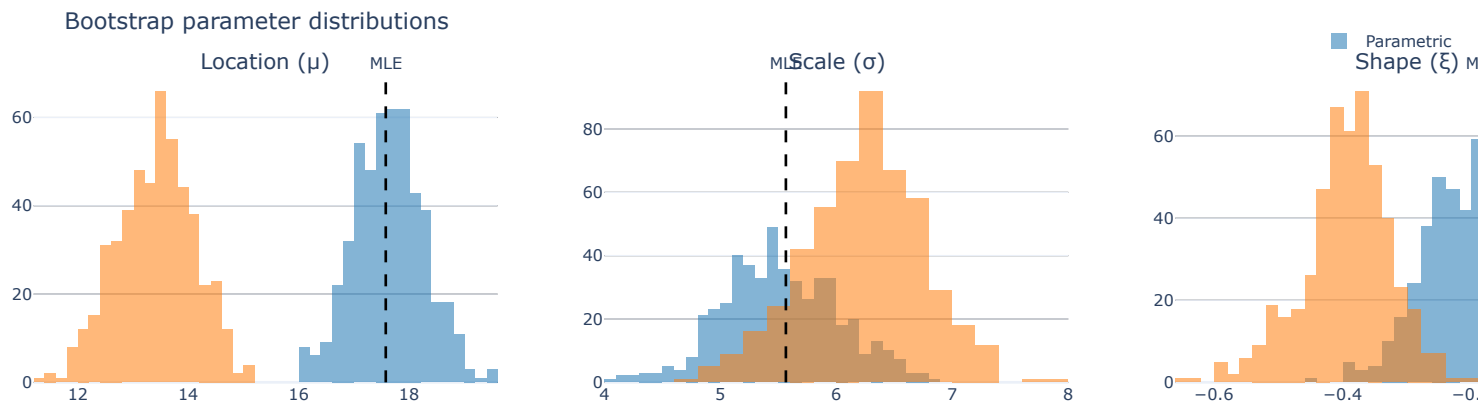
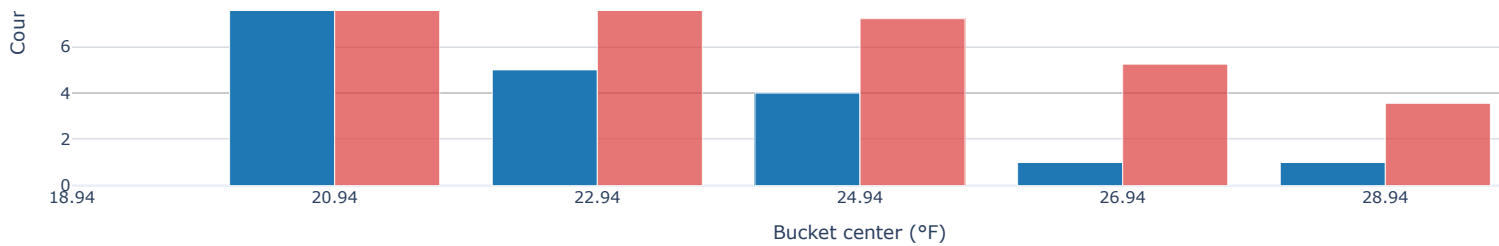


Bucket diagnostic (2.0-°F bins, full range)



Bucket diagnostic — upper tail ($\geq p75$)





► Parameters used